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Student Performance

Exploratory Data Analysis Report

10,000 Records · 14 Features · 5 Grade Levels

Prepared by: Data Analysis Division
May 2026

1. Executive Summary

This report presents a comprehensive exploratory data analysis (EDA) of a student performance dataset containing 10,000 records across 14 features. The dataset is complete with no missing values and is balanced across gender, internet access, and extra-class participation. The analysis surfaces key predictors of academic outcomes, grade distribution patterns, and actionable focus areas for institutional improvement.

Dataset Overview

Metric	Value	Notes
Total students	10,000	Complete records, no nulls
Features	14	8 numeric, 6 categorical
Mean overall score	63.83	SD = 10.23, Range: 33.3 – 96.9
Median score	63.78	Near-normal distribution
Score (25th pct)	56.53	Lower quartile boundary
Score (75th pct)	71.16	Upper quartile boundary
Grade levels	5	A, B, C, D, F
Dominant grade	C (50.7%)	5,073 of 10,000 students

Key Findings at a Glance

Final exam is the strongest predictor	Correlation $r = 0.689$ with overall score — nearly double the midterm's influence ($r = 0.529$).
Attendance separates grade bands clearly	A-grade students average 78.9% attendance vs 57.5% for F-grade — a 21-point gap.
Study hours show no performance signal	Correlation of -0.004 . All grade bands average ~5.4 hrs/day; raw time alone predicts nothing.
Grade C dominates the cohort	50.7% of students sit in grade C. Only 1.54% achieve A; 0.61% receive an F.
Gender, sleep, parent education are flat	Less than 0.4 score points separate any subgroup — these are non-factors in this dataset.

2. Grade & Score Distribution

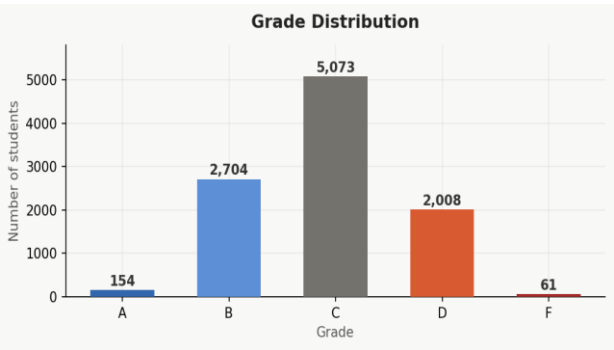


Fig 1. Number of students per grade

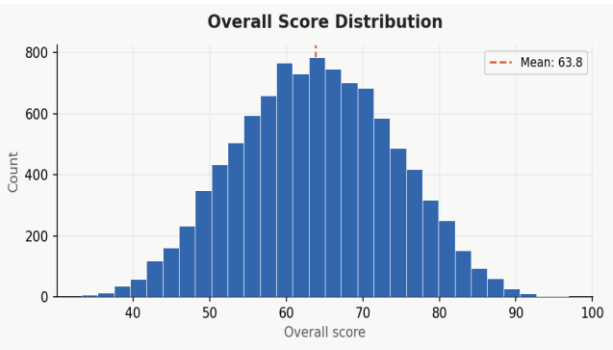


Fig 2. Histogram of overall scores

Grade Breakdown

Grade	Score Range	Count	% of Total	Avg Score
A	85.0 – 96.9	154	1.54%	87.6
B	70.0 – 84.9	2,704	27.04%	75.5
C	55.0 – 70.0	5,073	50.73%	62.7
D	40.1 – 55.0	2,008	20.08%	50.0
F	33.3 – 39.9	61	0.61%	37.8

The score distribution is approximately normal, centered at 63.83 (SD = 10.23). The clear majority of students (50.7%) land in the C band. The scarcity of A grades (1.54%) and the rarity of F grades (0.61%) suggests a well-calibrated assessment system with realistic differentiation in the mid-range.

3. Score Analysis by Grade

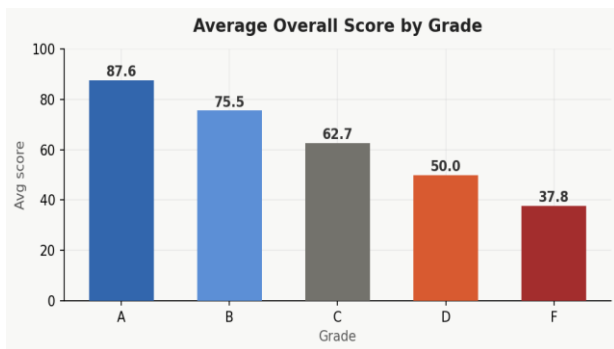


Fig 3. Average overall score by grade

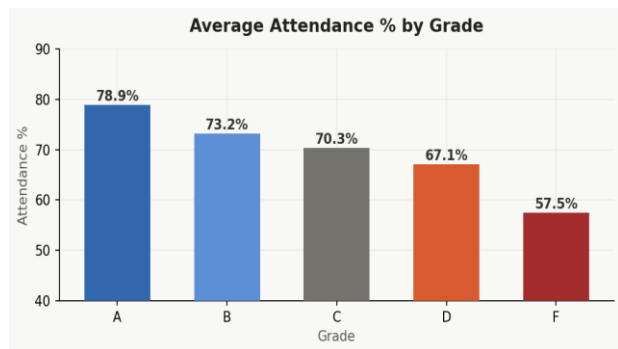


Fig 4. Average attendance % by grade

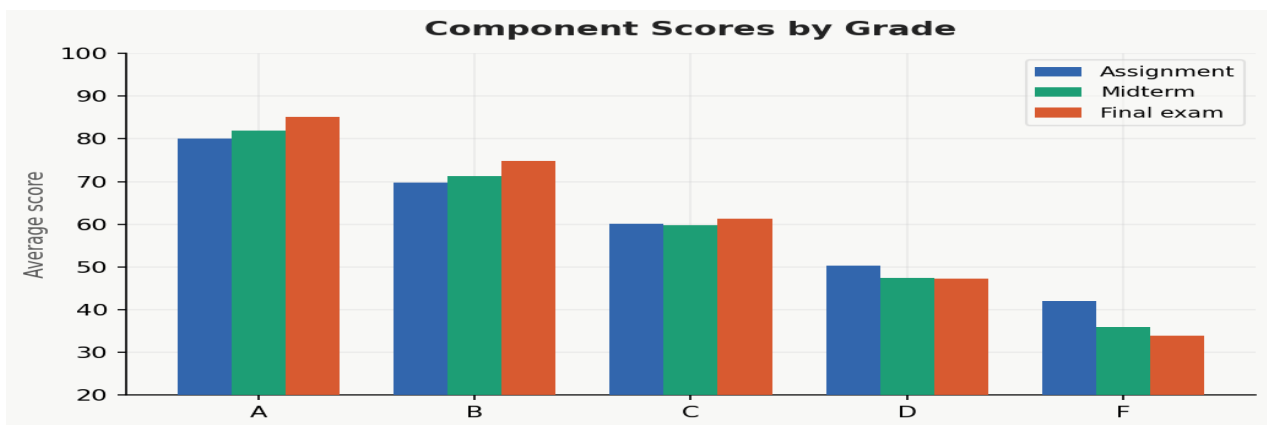


Fig 5. Assignment, midterm, and final exam scores by grade (grouped)

All three assessment components (assignment, midterm, final exam) scale cleanly with grade. The final exam shows the widest spread — a 51-point gap between A and F grades (85.2 vs 33.9) — underscoring its outsized weight in determining overall outcomes. Attendance follows a similarly clear gradient, with A-grade students attending 21.4 percentage points more than F-grade students.

4. Key Factors & Correlations

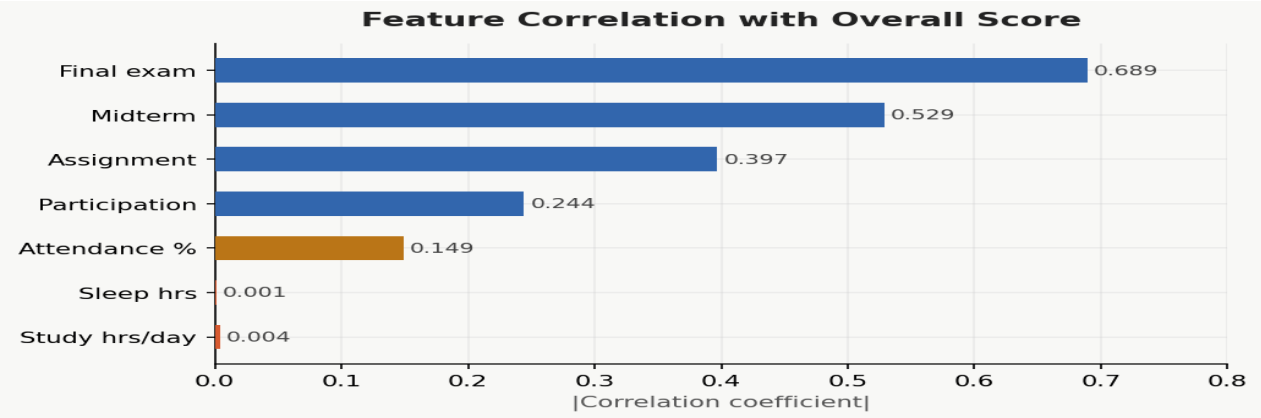


Fig 6. Absolute Pearson correlation of each feature with overall score

Feature	Correlation (r)	Strength	Direction
Final exam score	0.689	Strong	Positive
Midterm score	0.529	Moderate	Positive
Assignment score	0.397	Moderate	Positive
Participation score	0.244	Weak	Positive
Attendance %	0.149	Weak	Positive
Sleep hours	−0.001	Negligible	Negative
Study hrs/day	−0.004	Negligible	Negative

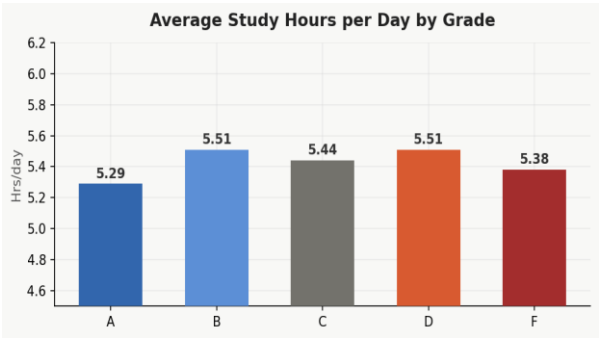


Fig 7. Average study hours per day by grade — all bands cluster at ~5.4 hrs

5. Demographic & Contextual Factors

Factor	Group	Avg Score	Delta
Gender	Female	63.83	—
	Male	63.82	−0.01
Internet access	Yes	64.12	—
	No	63.53	−0.59
Extra classes	Yes	63.89	—
	No	63.76	−0.13
Parent education	Bachelor	64.01	—
	Master	63.92	−0.09
	High School	63.75	−0.26
	PhD	63.62	−0.39

None of the demographic or contextual factors produce meaningful score differences. The largest observed gap — internet access (0.59 points) — is likely attributable to digital resource access rather than socioeconomic confounding. The dataset's balanced sampling (each subgroup near 50/50 or 25% splits) suggests deliberate stratification, which limits the ability to detect naturally occurring demographic effects.

6. Summary Insights & Recommendations

Priority	Area	Rationale
HIGH	Final exam preparation support	$r = 0.689$; single strongest predictor
HIGH	Attendance monitoring programs	21-pt gap between A and F grade cohorts
HIGH	Assignment completion tracking	$r = 0.397$; consistent upstream signal
MEDIUM	Participation incentives	$r = 0.244$; moderate signal worth reinforcing
LOW	Study hour mandates	$r = -0.004$; not supported by this data
LOW	Gender or sleep interventions	Essentially zero variance explained

This EDA confirms that academic performance in this dataset is driven primarily by formal assessment quality and consistent class attendance — not by background factors or raw study hours. Institutions aiming to improve outcomes should concentrate on examination readiness programmes, early-warning attendance systems, and structured assignment tracking. Demographic equity interventions, while important for other reasons, are not indicated as performance levers by this particular dataset.